

COSA DIRE ALLA PRESENTAZIONE

Slide 1

"Good morning, professors. Today we are presenting our project, '**Credit Risk Forecasting: A Machine Learning Approach**'.

In this presentation, we will guide you through our complete workflow. We will show you how we prepared the data, how we created a new variable to measure financial stress, and finally, how we compared two different predictive models.

Let's move to the next slide to look at our main objectives and the dataset."

Slide 2

[Transition from Slide 1] > "Let's move to the second slide, where we present our project objectives and an overview of the dataset."

[Part 1: Objective] "The main goal of our project is to build a predictive Credit Scoring system. Specifically, we want to predict the probability that a customer will default on their credit card payment next month.

To achieve this, we did a comparative analysis between two different models: a traditional Logistic Regression and a machine learning model called Random Forest.

Our core focus was not just running the models, but improving their performance by fixing two major issues in the data: multicollinearity—which means variables that are too correlated—and class imbalance."

[Part 2: Dataset] "Now, looking at the dataset: we used data from the famous UCI Machine Learning Repository. It has about 30,000 observations—or customers—and 24 features, which include demographic profiles, transaction history, and past payments.

A very important characteristic is the Class Imbalance: 78% of the customers are regular payers, while only 22% are in default and this is something we'll have to deal later.

Finally, we did some data cleaning. We found minor categories in the 'EDUCATION' and 'MARRIAGE' variables, so we regrouped them to make our models more stable."

Slide 3

[Transition & Hook] "Going more in depth with the variables, on the left side of this slide, we can see how we grouped them into four main blocks: Demographic Profile, Financial Capacity, Behavioral History, and Financial Exposure.

[Moving to the Plot] To immediately understand how these variables relate to our target, we isolated LIMIT_BAL—which represents the credit limit granted by the bank—and we created the boxplot you see on the right."

[Point 1: The Position of the Box] "There are two very important things to say about this graph. First, look at the position of the boxes. The central box—which represents the middle 50% of the customers—is much higher for the 'No Default' group than for the 'Default' group.

In business terms, this proves that the bank's initial screening works well. The bank gives high credit limits only to people it considers safe and wealthy. On the other hand, customers with low credit limits are naturally more fragile and statistically more likely to default.

[Point 2: The Median Line] "Second, look at the median line inside the boxes. For regular customers, the line is almost perfectly in the middle, which means the distribution is balanced. But for the default group, the median is squeezed down near the very bottom of the box.

Statistically, this means there is a huge concentration of data points at the very low values. In terms of business, this shows that default is mostly an entry-level phenomenon. The majority of people who do not pay are concentrated on the lowest credit cards. We can call this a 'floor effect': customers with minimal credit lines are the first to go into financial crisis as soon as they accumulate small debts."

Speech for Slide 4: Multivariate Analysis & Multicollinearity

[Introduction] "Now, let's look at slide number 4. This is a ggpairs plot.

For our Big Data project, this plot is a goldmine because it summarizes distributions, correlations, and scatter plots for our 5 key variables in a single view. Let's break it down by zones.

[1. The Diagonal - Density Plots] "First, let's look at the diagonal, which shows the shape of each variable.

- For **LIMIT_BAL**, the curves overlap significantly at the lower end. Statistically, this means a low credit limit alone cannot perfectly predict default, because the bank gives entry-level cards to many people who actually pay regularly. This high overlap is the exact reason why we needed feature engineering—like our **MAX_UTIL** indicator—to see how customers actually use their limit.
- For **PAY_0**, there is a clear split. Regular customers peak at zero or negative numbers, while the teal curve shifts to the right, showing a bump on positive values like 1, 2, or 3 months of delay. This is visual proof of why this variable is the most important feature in our machine learning models.

[2. The Upper Triangle - Correlation Coefficients] "Next, in the upper triangle, we can see the linear correlation numbers.

- **AGE** has a correlation very close to zero with all other variables. This proves that age provides independent information compared to financial data.
- **LIMIT_BAL vs PAY_0** shows a negative correlation around -0.25. This means that when the credit limit increases, payment delays decrease, which makes perfect sense for a bank.
- **LIMIT_BAL vs BILL_AMT1** has a moderate positive correlation of 0.28. This tells us that absolute bill amounts do not grow proportionally with the credit limit. A high bill might be safe for a premium customer but dangerous for a low-limit customer. This moderate correlation is exactly why we created our new variable, **MAX_UTIL**, to transform absolute debt into a relative financial stress indicator.

[3. The Lower Triangle - Scatter Plots] "Finally, the lower triangle shows the scatter plots.

- In the plots with **PAY_0**, we see a 'stripes' effect because repayment status moves in whole numbers like -1, 0, 1, 2.

- More importantly, we see a clear color segregation. In the plot between LIMIT_BAL and PAY_0, the top area—where delays are 2 months or more—is completely full of teal dots. This shows that if a customer is more than 2 months late, a high credit limit cannot save them from default."

Speech for Slide 5: Feature Engineering & The MAX_UTIL Indicator

[Part 1: Introduction and the "Why"] "Let's move to slide number 5, We realized that the original monthly bill data, expressed in absolute money values, could mislead our models.

For example, a 5,000-dollar debt has a completely different meaning if a customer has a 5,000-dollar credit limit or a 100,000-dollar credit limit.

We did that by creating an average for every month basically, dividing the bill of the month for the limit of the card and simply taking the highest over the 6 months.

[Part 2: Reading the Plot] "Looking at the two distributions, we can immediately see a clear behavioral difference between the two groups.

- The typical customer who pays regularly maintains a very prudent use of their card, with a median peak usage around **38%**.
- In contrast, for the group of customers who default, the median jumps vertically to **60%**, showing a much higher and more dangerous financial exposure during the six months.

[Part 3: The Final Insight] "But the most important and predictive insight comes from looking at the third quartile of the teal box, which aligns almost perfectly with the **1.0 mark**.

This gives us crucial evidence for the bank: exactly 25% of the customers who default completely maxed out or even over-drafted their credit limit at least once in the previous months. This strong separation between the two distributions explains why MAX_UTIL becomes one of the most significant and powerful predictors in both our Logistic Regression and Random Forest models."